

Survey on Information Requirements on the Google Books Ngram Corpus

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Abstract

The development of word frequencies over time is the subject of research in different branches of the humanities. Large temporal n-gram corpora have been created for this purpose, most notably the *Google Books Ngram Corpus*. While the concrete research questions vary between the different research works, there are similarities in the more abstract underlying information requirements, i.e., the structure of queries against a potential database system. Based on a systematic literature review, we extract these information requirements, leading to a categorization of existing articles into macro-areas of information requirements. Furthermore, we collect existing query systems for temporal n-gram corpora and evaluate their expressiveness regarding the information requirements we found.

Keywords: Google Books Ngram Corpus, Information Requirements, Literature Survey, Query Language.

1 Introduction

Analyzing the evolution of word frequencies in collections of written documents over time is a valuable tool often used, for instance, in the fields of linguistics ([Schlüter and Vetter 2020](#); [Breit 2017](#); [Davies 2014](#)), psychology ([Younes and Reips 2018](#); [O’Sullivan et al. 2019](#); [Teepe et al. 2023](#)), and philosophy ([Willkomm et al. 2018](#); [Schmidt-Petri](#)

et al. 2021). To help researchers access this information, a number of diachronic n-gram corpora have been created. These include the *Google Web 1T* corpus (Brants and Franz 2006) and the n-gram corpus of the Norwegian National Library (Birkenes et al. 2015). Among them, the Google Books Ngram Corpus (GBNC) (Lin et al. 2012; Mann et al. 2014; Michel et al. 2011) is by far the largest, comprising approximately 6% of all books ever published (Lin et al. 2012). For this reason and due to its vast influence, we focus on this specific corpus. The GBNC contains the yearly frequencies of billions of n-grams in several different languages. It was constructed from parts of the *Google Books* collection, with contributions from international libraries and universities. The exact composition of the corpus is, however, unknown.

Information Requirements.

When analyzing the GBNC, researchers use query systems to search for some specific kind of information, such as significant changes in word frequencies or correlations to real-world events. Understanding the researchers’ *information requirements* (or *information needs*) is therefore crucial for designing useful query systems (Taylor 1968; Owei 2003). However, the current literature on the topic is limited. A non-systematic, small-scale study on 11 publications has investigated which subsets of the GBNC are particularly interesting for researchers (Richter and Böhm 2024) and proposes a workflow to efficiently extract such subsets. In addition, Richter et al. (2025) examined existing query systems for temporal n-gram corpora with regards to their general expressiveness and usability. A similar effort is the *Archive Unleashed* project (Ruest et al. 2021), investigating how researchers from the humanities use web archives in general, with a focus on usability by non-technical users. Notable examples of query systems that are tailored to the *Digital Libraries* community are *LexiDB* (Coole et al. 2016, 2020) for corpus management and *SchenQL* (Kreutz et al. 2020, 2022) for metadata analysis.

A systematic literature review on information requirements regarding temporal n-gram corpora is still missing. This gap in the existing literature sparked the research questions for this paper:

- **Q1:** What kind of (implicit or explicit) information requirements do researchers have when working with the GBNC?
- **Q2:** Are existing query systems for the GBNC powerful enough to satisfy these information requirements?
- **Q3:** If not, what is missing in these query systems?

Contributions.

Our main contribution is an extensive, systematic literature review on applications of the GBNC in research. We survey, summarize and categorize 69 existing research articles and extract common information requirements from them. Furthermore, we discuss the ability of existing query systems to satisfy these information requirements.

Paper outline.

Section 2 introduces fundamental definitions and notations used in the following. In Section 3, we describe the process and results of our literature review. In Section 4,

existing query systems and their expressive power are discussed. Section 5 summarizes our findings.

2 Fundamentals

This section presents the basic concepts used in the next sections, largely following the definitions of Richter et al. (2025). We first describe temporal n-gram corpora (TNCs) and how to transform a text corpus into a TNC. Then, we define the term information requirements.

2.1 Temporal n-Gram Corpora

A temporal n-gram corpus (TNC) can be built from a temporal text corpus and contains information about the evolution of frequencies of n-grams (i.e., sequences of n words) in that corpus over time. For instance, the GBNC is based on the *Google Books* corpus (Michel et al. 2011). Before defining TNCs, we first fix an alphabet, i.e., a collection of symbols, which we indicate with Σ . The usual choice for Σ , in the context of query systems for the GBNC, is the set of English words, numbers and punctuation symbols, together with a wildcard symbol $*$. In this section, we assume that all the documents and n-grams are built from the same alphabet.

Temporal text corpora.

A *text document* is a semi-structured data object (Abiteboul 1997; Buneman 1997) $d = (c_d, m_d)$, where

- c_d is the *content* — a text written with symbols from Σ ;
- m_d is the *metadata* — a key-value mapping of properties of the text.

Given a metadata key k , we indicate with $d[k]$ the corresponding value — as a shorthand notation for $m_d(k)$. If k is not in the domain of m_d , i.e., it is not a valid key for m_d , we define $d[k] := \perp$.

A *text corpus* D is a set of text documents, $D = \{d_1, \dots, d_l\}$. A text corpus is *temporal* if the metadata of its documents contains time-related information. More formally, we say that D is temporal if there exists some timestamp key k_t such that, for any document $d_i \in D$, $d_i[k_t]$ is a timestamp. This timestamp key can, for instance, be ‘publication date’.

Temporal n-gram corpora (TNC).

An *n-gram* $\mathbf{g} = (\sigma_1, \dots, \sigma_n) \in \Sigma^n$ is an ordered tuple of symbols belonging to Σ . The frequency an n-gram $\mathbf{g}$ *occurs* in a document d is the number of times the concatenated symbols $\sigma_1 \dots \sigma_n$ (ignoring whitespace characters) appear in c_d . The *frequency function* $f_{\mathbf{g}} : T \mapsto n \in \mathbb{N}_0$ associates to a timestamp T the total number of occurrences of $\mathbf{g}$ within all documents in D with timestamp T . W.l.o.g., we also allow $f_{\mathbf{g}}$ to take as input an ordered tuple of timestamps $\mathbf{T}$. In this case, $f_{\mathbf{g}}$ returns an ordered tuple of frequencies.

We can finally define a temporal n-gram corpus, which contains temporal information on the frequency of n-grams over a certain time span.

Definition 1 (Temporal n-gram corpus (TNC) (Richter et al. 2025)) A *temporal n-gram corpus* is a tuple $C = (G, \mathbf{T}, F)$, where G is a set of n-grams, $\mathbf{T} = (T_1, \dots, T_m)$ is an ordered tuple of unique timestamps, and $F = (f_{\mathbf{g}})_{\mathbf{g} \in G}$ is a family of frequency functions, containing one frequency function per n-gram.

2.2 Information Requirements

Information requirements, also known as information needs, are a key component to the design of query systems (Wilson 1981; Taylor 1968). In this paper, by information requirement we mean a *formalized need*, i.e., a rigorous and precise description of the user’s conscious information need (Taylor 1968). A simple information requirement regarding the GBNC could be, for example, ‘*What is the most frequent n-gram in the year 2017?*’. Users typically try to satisfy their information requirements with a query system. As a consequence, query systems should be designed to address the specific information requirements of their potential users (Wilson 1981).

3 Survey of Information Requirements for the GBNC

The goal of our survey is to answer the first of our research questions, **Q1**: What kind of (implicit or explicit) information requirements do researchers have when working with the GBNC? **Q2** and **Q3** are further discussed in Section 4.

First, we present the methodology of the survey, including our criteria for literature selection. Then, we classify papers according to how they make use of the GBNC. Finally, we discuss and categorize the information requirements we identified.

3.1 Survey Methodology

We conduct a systematic literature survey, following accepted guidelines for such studies (Kitchenham 2004; Moher et al. 2010; Page et al. 2021) and aim to make our work reproducible by establishing a survey procedure as well as rigorous, objective literature selection criteria. We publish all the relevant code and data.¹

3.1.1 Literature Selection

Over the years, the GBNC has been the subject of a substantial amount of studies. For instance and just using the Google Scholar search engine with specific keywords,² we could easily retrieve about 1000 distinct publications. The number of retrieved publications increases by more than one order of magnitude when including alternate spellings and/or orderings of the keywords.

A major obstacle for our survey is the fact that most of these publications mention the GBNC only as related work or as a side note. These publications are not relevant for our research question, as they do not pose any information requirements towards

¹https://github.com/DrCohomology/GBNC_Survey

²[https://scholar.google.com/scholar?q=\"google+books+ngram+corpus\"](https://scholar.google.com/scholar?q=\), last accessed August 26, 2025.

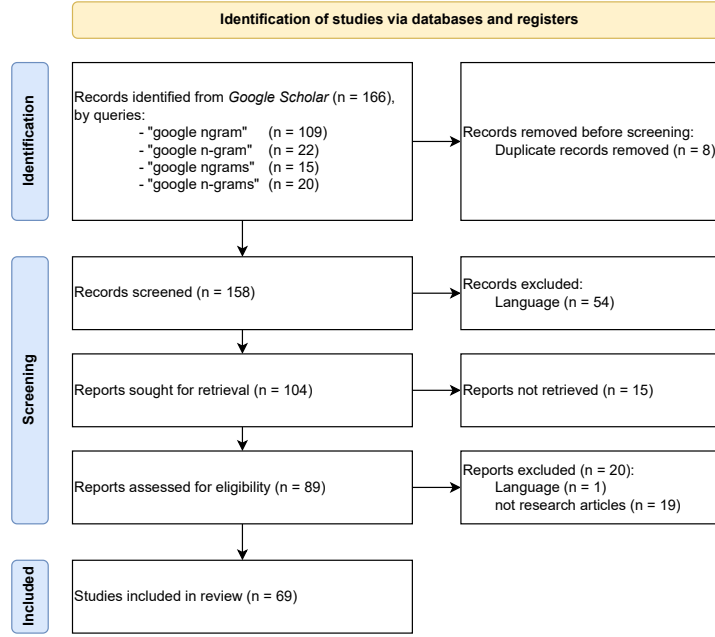


Fig. 1 The *PRISMA 2020* (Page et al. 2021) flow chart of our literature selection procedure

the GBNC. To filter out such works and to focus on articles that work directly with the GBNC, we consider only publications that contain **google ngram**³ in the title.

Our reasoning for choosing this specific set of queries is as follows: Mentioning a dataset in the title of a research article indicates that this specific dataset plays an important role in the article. We acknowledge that the inverse is not always true, especially when a dataset is known under different names. What we call ‘Google Books Ngram Corpus’ is also referred to as ‘Google Ngram Corpus’ or ‘Google Books Corpus’ in literature. The first case is covered by our queries, the second one is not. We have considered, but ultimately rejected, other queries as well: Works that have **google books** but not **ngram** in the title usually refer to the broader *Google Books* project, including full texts and metadata, and their number is too large for manual screening. Further restriction to works including **google books corpus** but not **ngram** in their title yields only 11 results, none of which would pass all other selection criteria.

While manual screening of all papers is, when feasible, preferable, we deemed this automatic solution necessary to obtain an impartial and reproducible survey while keeping screening efforts manageable. Furthermore, with this procedure, we intended to include more publications that focus on working with TNCs—rather than just mentioning them.

³or slight spelling variations, namely: **google n-gram**, **google ngrams**, and **google n-grams**; with all queries being case-insensitive

Thus, 166 results remain, 8 of which were duplicates. We then excluded 55 more papers written in languages other than English; 19 that are blog posts, presentations, or book chapters; and 15 papers of which we could not retrieve the full text. This leaves a total of 69 articles to be considered, the *PRISMA 2020* (Page et al. 2021) flow chart of this procedure is presented in Figure 1. In some older publications within this set of 69 articles, *Google Ngram Corpus* refers to the *Google Web 1T corpus* (Brants and Franz 2006) instead of the GBNC. Since the two corpora are structurally similar, we decided to include these publications nonetheless.

3.1.2 Goals and Procedure

The core goal of our survey is to answer **Q1**: What kind of (implicit or explicit) information requirements do researchers have when working with the GBNC? A secondary goal is to identify patterns within these requirements, for example by examining which types of studies have which information requirements. We therefore perform two orthogonal categorizations: grouping of publications with regard to their usage of the GBNC (cf. Section 3.2.1), and grouping of information requirements into macro-areas (cf. Section 3.2.2).

The categorization procedure was as follows: First, author A1 surveyed an initial set of 30 publications, selected at random. From these publications, A1 extracted the concrete information requirements and assigned them to their respective macro-areas. During this phase, A1 also fixed the four usage groups and assigned the respective articles to them, reserving the option to add more groups if necessary. Second, author A2 then repeated the analysis on another set of 20 publications, assigning publications to groups and information requirements to macro-areas. Differences in nomenclature of information requirements were then resolved by A1 and A2 together. The remaining 19 publications were again assigned to A1, who applied the same procedure as before, the results were checked by A2 afterwards.

Details regarding the statistical analyses in Section 3.3.1 are discussed there.

3.2 Survey Overview

We classify the publications according to how they use the GBNC and aggregate the information requirements into macro-areas.

3.2.1 Usages of the GBNC

We have observed that the 69 investigated research articles can be grouped into four distinct usage groups. This grouping is based on the viewpoint the respective works take towards the GBNC:

- Some researchers treat the GBNC as *data*, for example by using it to train statistical models. In these works, the GBNC is used as a tool, not as a subject of research in itself.
- In other articles, the GBNC is *analyzed* in its own right. Here, researchers try to understand the inner workings of the GBNC, for example regarding its composition or statistical anomalies within the corpus.

- Several *query systems* have been proposed, aiming to make analyses of the GBNC and similar corpora easier.
- Finally, some articles introduce *new corpora*, similar to the GBNC, but in different languages or tailored to specific domains, for example by only including scientific texts, as in the work of Chumtong and Kaldewey (2017).

Table 1 Types of information requirements

Macro-area	Information requirements	
Frequency plots	frequency plots	
n-Gram filtering	specific n-grams search PoS search	wildcard search inflection search
Frequency filtering	similarity between frequencies first/last occurrence	peaks and valleys of an f_g
External knowledge	real-world data other corpora	dictionaries
Natural Language Processing	sentiment analysis named entity recognition topic grouping	collocates word embeddings
Statistical computations	change point detection trend prediction entropy	dependency analysis similarity between frequencies n-grams clustering

3.2.2 Categorization of Information Requirements

We distinguish six macro-areas of information requirements. Each of these macro-areas contains a finer-grained set of information requirements. The categorization is summarized in Table 1.

Frequency plots.

Plotting the frequencies of specific n-grams is the main requirement satisfied by the Google Books Ngram Viewer (Michel et al. 2011; Mann et al. 2014; Lin et al. 2012). These plots are often used to visually identify and explore trends or dependencies between frequencies.

n-gram filtering.

n-gram filtering deals with the search of n-grams matching a specific pattern. We identified *specific n-grams search*, *wildcard search*, *part-of-speech (PoS) search*, and *inflection search* within the investigated publications.

Frequency filtering.

Frequency filtering is related to filtering for specific n-grams g whose frequency function f_g satisfies certain criteria. The criteria we found are: *similarity between frequency functions*, *peaks/valleys of the frequency function to occur in certain time periods*, and *first/last occurrence of an n-gram*.

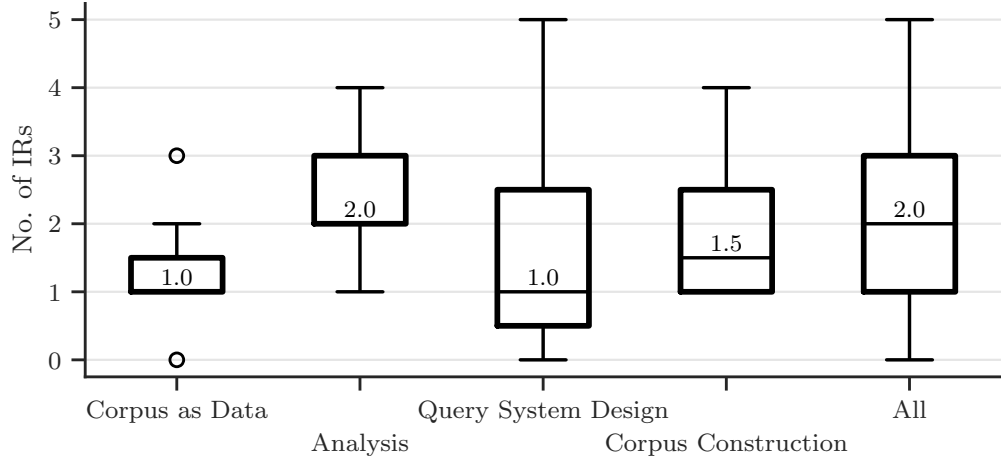


Fig. 2 Box plots for the number of information requirements per group of publications

External knowledge.

Sometimes, the information within the GBNC is not enough to satisfy the conscious need of the users. In this case, they might add external knowledge to their query, for example to relate real-world data like stock prices or literacy rates to frequencies within the corpus. To handle this, it is important that the query system can easily consider data from different sources. We found the following most common external knowledge bases: *real-world data*, *dictionaries* (such as WordNet (Miller 1995; Fellbaum 1998)), and *other corpora* (such as the Corpus of Historical American English (Davies 2012)).

Natural language processing (NLP).

This macro-area encompasses all information requirements related to analyzing or processing natural language. This includes *sentiment analysis*, *collocates*, *named entity recognition*, *word embeddings*, *topic grouping*.

Statistical computations.

Finally, the user’s conscious need might require a more sophisticated statistical analysis of the n-gram frequencies. We identified *change point detection*, *dependency analysis*, *trend prediction*, *similarity between frequency distributions*, *entropy*, and *clustering of n-grams*.

3.3 Information Requirements on the GBNC

In the following, we first describe different applications of the GBNC in the 69 articles we investigated. Afterwards, we discuss their information requirements according to our macro-areas proposed earlier.

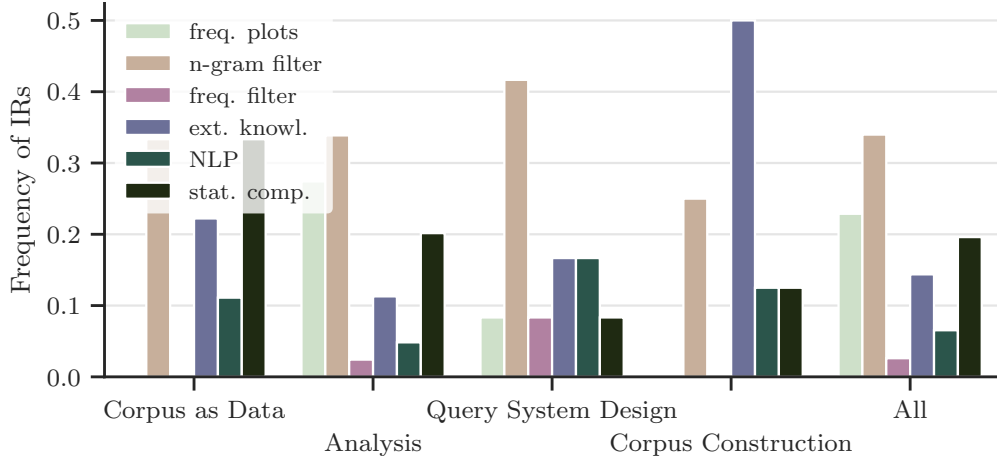


Fig. 3 Distribution of macro-areas of information requirements across the groups of publications

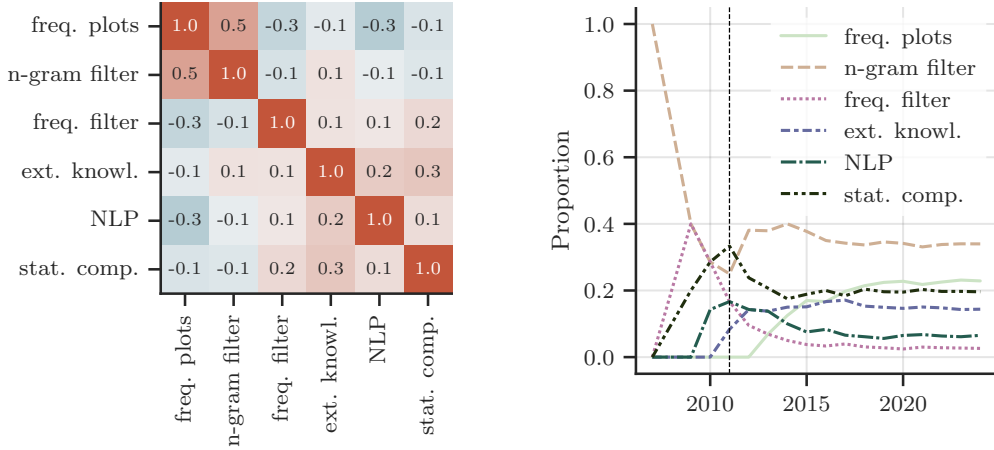


Fig. 4 Correlation between macro-areas of information requirements.

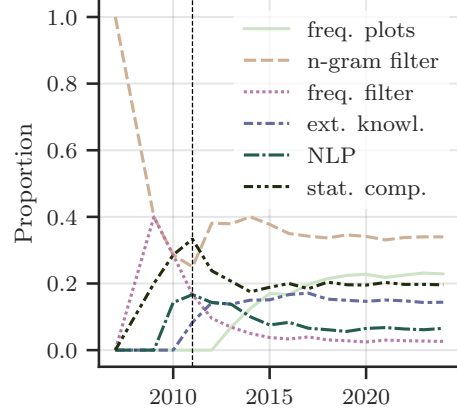


Fig. 5 Proportion of information requirements over the years. The GBNC and GBNV were published in 2011 (Michel et al. 2011).

3.3.1 Overview of Results

Table 2 summarizes our findings, distinguishing the usage groups of publications and macro-areas of information requirements, as introduced in Section 3.2. Figure 2 shows a box plot for the number of information requirements per publication, broken down by the usage groups described in Section 3.2.1. Figure 3 further illustrates how the frequency of different macro-areas of information requirements introduced in Section 3.2.2 varies across the different groups of publications. The majority of the publications we surveyed mention between one and three information requirements.

The most frequent information requirements are n-gram filtering, frequency plots, and statistical computations, while frequency filtering is the least common.

The numbers and types of information requirements per publication vary with the usage group of the publication. Publications using the GBNC as data have, in general, the least amount of information requirements. The most common ones in this group are n-gram filtering, statistical computations, and integration of external knowledge. Frequency plots are notably absent from this group—which is to be expected, as frequency plots are mostly a tool for visual data exploration to analyze the GBNC itself. Publications analyzing the GBNC typically have at least two information requirements, most often frequency plots and n-gram filtering. Only a small number of these publications need frequency filtering. Publications which design a new query system usually consider between one and two information requirements. Almost half of them include n-gram filtering, while only one includes statistical computations. The other information requirements are fairly evenly distributed. Finally, publications that use the GBNC to build another corpus have between zero and four information requirements. 3 out of 4 publications need to incorporate external information, e.g., external text collections on which TNCs are constructed or specific dictionaries to restrict the GBNC to smaller subsets.

Figure 4 shows that frequency plots and n-gram filtering tend to co-occur together very often. A frequency plot can only display a limited number of time series while still being readable, so filtering is a necessary step for plotting. The focus on n-gram filtering instead of frequency filtering might be because n-gram filtering is much better supported by query systems, as detailed in Section 4. On the other hand, frequency plots and NLP are (weakly) mutually exclusive. One possible explanation for this is that most publications with frequency plots analyze the GBNC itself, which contains only fragments of natural language. Not all NLP methods are applicable to such fragments. In other types of publications, a few suitable NLP methods are nonetheless applied to these language fragments, or to external texts, but these works then usually do not rely on frequency plots.

Finally, Figure 5 shows the proportion of the different information requirements over the years. It is notable that frequency plots start to be used in 2013, after the introduction of the *Google Books Ngram Viewer* (Michel et al. 2011) in 2011. Similarly, n-gram filtering gained popularity over the years and is the information requirement mentioned most frequently since 2012. A contrary effect can be seen for frequency filtering and statistical computations, two types of information requirements the *Google Books Ngram Viewer* does not support at all.

In the following, we detail on some of the major findings on our survey of information requirements within each of the usage groups of the GBNC. Here, publications are not grouped by their information requirements, but by their perspective on the GBNC and the specific tasks they tackle. While correlations between the two systems of categorization exist (see Fig. 3), they are, in principle, orthogonal to each other.

3.3.2 Corpus as Data

In several of the investigated works, the GBNC is used as training data for statistical models, or as a generic benchmark dataset.

Table 2 Overview of the major results of our survey

Usage of GBNC	Reference	Main Information Requirements					
		freq. plots	n-gram filter	freq. filter	ext. knowl.	NLP	stat. comp.
Corpus as Data	Basile et al. (2016)				✓		✓
	Bhowmick et al. (2015)						
	Bochkarev et al. (2016)						✓
	Nazar and Renau (2012)		✓				
	Perrie et al. (2013)		✓		✓	✓	
	Savinkov et al. (2021)						✓
	Yu et al. (2007)		✓				
Analysis	Bai and Huang (2018)	✓	✓				✓
	Bochkarev et al. (2020a)					✓	
	Bochkarev et al. (2020b)		✓				✓
	Bochkarev et al. (2018)						✓
	Brandt (2018)	✓	✓				
	Bukhtoyarov and Bukhtoyarova (2018)	✓	✓		✓		
	Caruana-Galizia (2016)	✓	✓		✓		✓
	Chen and Huang (2022)	✓	✓		✓		
	Chiru and Dinu (2017)			✓	✓		✓
	Clark et al. (2022)	✓	✓				
	Coraci et al. (2020)	✓					
	Dönmez (2018)	✓	✓				✓
	El-Ebshihy et al. (2018)	✓	✓			✓	✓
	Friginal et al. (2014)	✓	✓		✓		✓
	Galeev and Solovyev (2018)	✓	✓				
	Grant and Walsh (2015)	✓	✓				✓
	Guan et al. (2024)		✓		✓	✓	✓
	Gulordava and Baroni (2011)						✓
	Hill and Simha (2016)					✓	
	Inkpen and Islam (2010)					✓	✓
	Islam and Inkpen (2009)		✓	✓			
	Islam et al. (2012)		✓		✓		✓
	Joubarne and Inkpen (2011)		✓		✓	✓	✓
	Juola (2013)						✓
	Klein and Nelson (2009)			✓			✓
	Koplenig (2017)	✓	✓				✓
	Lin et al. (2012)		✓				
	Macedo (2013)	✓	✓				
	Madsen and Slåtten (2022)	✓	✓				
	Moskovkin et al. (2019)	✓	✓				
	Nestik et al. (2022)	✓	✓		✓		✓
	Olimid et al. (2023)	✓	✓				
	Olimid et al. (2024)	✓	✓				
	O'Sullivan et al. (2019)	✓	✓				
	Pekina et al. (2018)		✓				✓
	Radimský (2022)		✓				
	Ribeiro and Lima (2017)	✓	✓				
	Richey and Taylor (2020)	✓	✓		✓		✓
	Roth (2013)	✓	✓				
	Roth (2014)	✓	✓				
	Roth et al. (2017)	✓	✓				
	Sampsel (2021)	✓	✓		✓		✓
	Skrebyte et al. (2016)	✓	✓		✓		
	Solovyev et al. (2019)	✓	✓		✓		
	Sparavigna and Marazzato (2015)	✓	✓				
	Teepe et al. (2023)	✓	✓				✓
	Younes and Reips (2018)	✓	✓		✓		✓
	Younes and Reips (2019)	✓	✓				✓
	Zeng and Greenfield (2015)	✓	✓		✓		✓
	Zięba (2018)	✓	✓				
	Zyukina et al. (2020)	✓	✓				
Query System Design	Aleksandrov and Strapparava (2012)		✓		✓	✓	
	Davies (2014)		✓				
	Grabowski and Swacha (2012)		✓				
	Habeeb et al. (2020)						
	Mann et al. (2014)	✓	✓				
	Schlüter and Vetter (2020)						
	Schmidt-Petri et al. (2021)		✓	✓	✓	✓	✓
Corpus Construction	Alsmadi and Zarour (2018)				✓		
	Chumtong and Kaldewey (2017)				✓		
	Ivanov (2014)		✓		✓		
	Ivanov and Solovyev (2020)		✓		✓	✓	✓
Total		35	52	4	22	10	30

Statistical models.

Basile et al. (2016) use the GBNC to train vector-embedding models and to detect temporal shifts in word meanings using these models. Nazar and Renau (2012) train a frequency-based grammar checker on the GBNC data. Savinkov et al. (2021) train a neural network on the GBNC to improve part-of-speech and case tagging in Russian texts. Yu et al. (2007) use the n-gram data from a smaller corpus to derive semantic relations between words.

Other applications.

Bhowmick et al. (2015) use the GBNC as a benchmark dataset for *Apache Hadoop* and *Apache Pig*.⁴ Bochkarev et al. (2016) use the GBNC to verify Heap’s Law, an empirical observation relating the number of unique words within a text to its length.

3.3.3 Analysis

The largest group of publications in our review considers the analysis of the GBNC itself as their subject of research.

Linguistics.

Temporal n-gram corpora track the development of n-gram frequencies over time, so the detection of syntactic and semantic shifts is one of their natural applications. Galeev and Solovyev (2018) examine the popularity of different verb forms over time to detect syntactic changes. In the work of Pekina et al. (2018), the GBNC is used for probabilistic estimates of vocabulary sizes and to detect the appearance of new words in a language. Chen and Huang (2022) use co-occurrence information from the GBNC to determine semantic change of a specific Chinese word. Similarly, Gulordava and Baroni (2011) use the distributional similarity of n-grams to identify semantic changes. El-Ebshihy et al. (2018) train word embeddings on the GBNC and apply change point detection to identify ‘interesting’ n-grams and time series. Koplenig (2017) investigates the adverse effects that the lack of metadata within the GBNC can have on such change detection tasks, e.g., due to potential undetectable imbalances in corpus composition.

Joubarne and Inkpen (2011) compare the semantic similarity of terms annotated by human experts to values automatically constructed from the GBNC. A similar effort is made by Islam et al. (2012), who compare different similarity measures based on the GBNC data.

Inkpen and Islam (2010) and Islam and Inkpen (2009) use a temporal n-gram corpus to derive spelling correction rules. Hill and Simha (2016) generate ‘fill-in-the-blanks’ questions from the GBNC data to assist with contextual language learning.

Lin et al. (2012) introduce syntactic annotations to the GBNC, mainly dependency parsing results. Bochkarev et al. (2018) use these to estimate the dynamic complexity of both English and Russian texts over time. Pekina et al. (2018) conduct a similar study based only on the estimated vocabulary sizes.

⁴<https://hadoop.apache.org>, <https://pig.apache.org/>

Radimsky (2022) uses the GBNC to construct a morphology database of Romance languages, while Bochkarev et al. (2020a) use the GBNC data for *Named Entity Recognition*, a standard task of natural language processing.

Other Scientific Disciplines.

Not only linguistic research is performed on the GBNC, researchers from other disciplines are interested in the data as well. Most frequently, authors from these other disciplines focus on a small set of n-grams within the GBNC, specific to their respective field and research question.

Bukhtoyarov and Bukhtoyarova (2018) investigate the relationship between the frequency of terms related to ‘revolution’ to actual historical revolutions. Their study considers five different languages and a small set of 1-grams in each of these. Macedo (2013) conducts a similar study on the history of ‘eurocentrism’. Olimid et al. (2024) consider a list of n-grams and their frequencies as a proxy indicator for the degree of participation of women in political decision making. Olimid et al. (2023) apply the same methodology on terms concerned with governance styles and democracy. Juola (2013) uses the GBNC to show that culture in the USA has grown more and more complex over time. Zięba (2018) gives a general overview of the GBNC in socio-cultural research. Younes and Reips (2019) present guidelines for improving the quality of GBNC research, using religious terms as a demonstration.

Sparavigna and Marazzato (2015) and Brandt (2018) look at terms from specific disciplines, i.e., material science and geology, to draw conclusions regarding the history of the fields. Similar studies analyze the history of architecture (Bai and Huang 2018), climatology (Grant and Walsh 2015), and education (Moskovkin et al. 2019; Sampsel 2021). Ribeiro and Lima (2017) investigate the popularity of certain food idioms over time. In a similar fashion, Dönmez (2018) and Zyukina et al. (2020) analyze the popularity of various human activities.

Bochkarev et al. (2020b) and Guan et al. (2024) investigate the development of color terms within the Russian and English GBNC, respectively. Bochkarev et al. (2020b) look at the prominence of two Russian terms for ‘brown’, while Guan et al. (2024) link the frequencies of different colors to psychological effects. Further studies on links from psychiatry and psychology to n-gram frequencies have been conducted by O’Sullivan et al. (2019) (history of psychiatry) and Teepe et al. (2023) (mental health and digitalization). Coraci et al. (2020) investigate the impact of diagnostic ultrasound, again by using specific n-gram frequencies, comparing the results between the GBNC and *PubMed*,⁵ a database of publications in medicine.

Madsen and Slåtten (2022) study the evolution of different emerging technologies and how they are mentioned in literature, e.g., ‘Big Data’ technologies or management concepts such as ‘Total Quality Management’, by looking at their respective frequencies in the GBNC. Roth (2013, 2014); Roth et al. (2017) apply the same methodology to functional differentiation.

⁵<https://pubmed.ncbi.nlm.nih.gov/>

Real-World Indicators.

Several works attempt to relate n-gram frequencies from the GBNC to real-world data. Zeng and Greenfield (2015) use “urban population, household consumption per capita [...] and tertiary school enrollment” in China as proxy indicators for sociodemographic developments. They compute the correlation between these indicators and the frequencies of terms related to individualism and collectivism. Similar studies for Russia are conducted by Skrebyte et al. (2016) and Nestik et al. (2022), for German-speaking countries by Younes and Reips (2018), and for Germany during the times of the Third Reich by Caruana-Galizia (2016).

Frequency Analysis.

Some publications analyze the n-gram frequencies themselves. Chiru and Dinu (2017) show that some n-grams have cyclic frequency patterns. Clark et al. (2022) search for n-grams with particularly strong trends in the Chinese GBNC. Frigal et al. (2014) compare the GBNC frequencies to n-gram frequencies generated from the *Corpus of Historical American English* (Davies 2012), while Solovyev et al. (2019) compare the GBNC to the *Russian National Corpus*⁶ in a similar fashion. Klein and Nelson (2009) investigate, in different TNCs, the correlation between term counts, i.e., the total number of occurrences of a term, and document frequencies, i.e., the number of documents said term appeared in.

3.3.4 Query Systems

Existing TNC query systems are discussed in more detail in Section 4. Here, we briefly discuss the works that also discuss particular information requirements and were thus part of our literature survey, which the vast majority of systems was not.

Query System Implementations.

The original query system for the GBNC is the *Google Books Ngram Viewer*⁷ (GBNV) (Michel et al. 2011; Lin et al. 2012; Mann et al. 2014). It allows users to input specific n-grams and retrieve their relative frequencies as plots. Davies (2014) presents an improved version of the GBNV, allowing for more flexible wildcard searches. Another graphical search interface is presented by Schlüter and Vetter (2020). This interface focuses on a specific linguistic phenomenon, namely the usage of ‘a’ or ‘an’ before words beginning with the letter ‘h’.

Query Languages.

Aleksandrov and Strapparava (2012) introduce *NgramQuery*, a query language that connects temporal n-gram corpora to *WordNet* (Fellbaum 1998; Miller 1995). This allows users to perform queries about semantic relationships, such as retrieving synonyms of a given word. The *Conceptual History Query Language* (Schmidt-Petri

⁶<https://ruscorpora.ru/>

⁷<https://books.google.com/ngrams/>

et al. 2021; Willkomm et al. 2018) introduces several operators to represent information requirements from *Conceptual History* (Koselleck 1978). Examples of this are an operator for topic grouping and an operator for searching similar frequencies.

Foundational Work.

Grabowski and Swacha (2012) present a custom encoding and compression scheme for the GBNC data. The encoding reduces file sizes compared to the files published by Google, so that the GBNC can be searched more efficiently and that query execution times are reduced. Habeeb et al. (2020) describe how to unzip data from the raw corpus for further processing.

3.3.5 Corpus Construction

Alsmadi and Zarour (2018) propose extending the GBNC by including Arabic resources. Chumtong and Kaldewey (2017) construct a TNC out of scientific publications. Ivanov (2014) and Ivanov and Solovyev (2020) use the GBNC to construct corpora of semantic knowledge and relationships between words. In all of these works, external knowledge is used, usually in the form of additional texts.

3.4 Main Survey Findings

The main result of our survey is the categorization of information requirements. Additionally, we report the following findings regarding the research questions posed in Section 1:

Regarding **Q1**, we find that *the concrete research questions vary between fields, but the underlying, more abstract information requirements are very similar*. Nearly every publication restricts the GBNC to a specific subset of interest, usually a set of n-grams related to the respective research field, in line with the findings of Richter and Böhm (2024). The integration of external knowledge, for example dictionaries or semantic ontologies, is necessary for many different applications of the GBNC. Similarly, statistical computations on n-gram frequencies play an important role in analyzing the GBNC. A query system that allows users to specify restricted subsets of the GBNC and supports these information requirements could therefore be used by researchers across different domains.

Our survey also gives some insight into questions **Q2** and **Q3**, with a more extensive examination of query systems following in Section 4: *Most of the existing query systems that were part of the survey itself only support one or two of the information requirement categories we found*. Especially, support for the integration of external knowledge and statistical computations is rarely available, and thus requires researchers to implement custom solutions.

There is a clear connection between the query systems’ capabilities and the popularity of different information requirements. One example for this is the significant under-representation of frequency filtering, which might be explained by a lack of support in most query systems.

4 Existing Query Systems

The survey presented in Section 3 introduces a set of information requirements that are common throughout literature. In the following, we examine existing TNC query systems with regard to their abilities of satisfying these information requirements.

Different query system for TNCs exist, a broader summary can be found in Richter et al. (2025). In that study, the capabilities of TNC query systems were evaluated from a different point of view, i.e., it is mostly descriptive in the sense that it evaluates what the systems *can* do. On the contrary, the present work in this paper focuses on the more normative aspect of what the systems *should be able to* do in order to answer research questions from literature, as given by the information requirements in Section 3.

Table 3 Existing TNC query systems

Query System	Information Requirements					
	freq. plots	n-gram filt.	freq. filt.	ext. knowl.	NLP	stat. comp.
Ngram Search Engine (Sekine and Dalwani 2010)		✓	✓			
GBNV (Mann et al. 2014; Michel et al. 2011; Lin et al. 2012)	✓	✓				
NgramQuery (Aleksandrov and Strapparava 2012)		✓		✓	✓	
PoliticalMashup Ngramviewer (de Goede et al. 2013)	✓	✓				
GB-Adv (Davies 2014)		✓				
Slash/A (Todorova et al. 2014)	✓	✓	✓			
NB N-gram (Birkenes et al. 2015)	✓	✓				
CHQL (Schmidt-Petri et al. 2021; Willkomm et al. 2018)		✓	✓	✓	✓	✓
(an:a)-lyzer (Schlüter and Vetter 2020)						
Icelandic Gigaword n-Gram Viewer (Steingrímsson et al. 2020)	✓	✓				
Meta Trend Viewer (Indig et al. 2022)	✓	✓			(✓) ⁸	
ParlaMint Ngram Viewer (de Jong et al. 2024)	✓	✓		✓		

The groups of information requirements we identified in Section 3 overlap with those found by Richter et al. (2025), but are not exactly equal. Richter et al. do not consider frequency plots as distinct information requirement, and thus do not list it in their review results. Our groups *n-gram filtering* and *frequency filtering* are contained in their *n-gram operations* and *filtering operations*, respectively. Furthermore, our more fine-grained distinction between the groups *external knowledge*, *NLP* and *statistical computations* is subsumed in their notion of *knowledge*, *metadata* and partially in *frequency operations*. Applying these correspondences to the results of Richter et al. (2025) yields Table 3. A checkmark in Table 3 represents the ability of a query system to satisfy *at least one* of the information requirements in the respective group.

While the internal architecture of the existing query systems differ, most systems follow a very similar usage pattern: The user specifies a short list of n-grams, potentially with some support for wildcards, the system then returns a line plot of their frequencies over time. For a more detailed view, we adopt the conceptual grouping of query systems established by Richter et al. (2025): The *GBNV-like* systems (GBNV, Ngram Search Engine, GB-Adv, Slash/A, NB N-Gram, Icelandic Gigaword n-Gram Viewer) mainly differ in their wildcard support and the underlying corpora. The

⁸Support is planned, according to the authors.

subcorpus-oriented systems (PoliticalMashup Ngramviewer, Meta Trend Viewer, ParlaMint Ngram Viewer) offer an additional filtering step: Based on metadata for their specific corpora, they allow a restriction of documents, for example by author or publication source. After this step, their usage is similar to the GBNV-like systems. The three remaining systems are more diverse: The (an:a)-lyzer examines a very specific linguistic phenomenon over time, with user-controllable parameters for the distinction between *UK* and *US* English, and the selection of specific, pre-defined groups of words to plot. NgramQuery and CHQL have no available implementations, but both introduce complex query languages to express information requirements against both TNCs and external knowledge, returning datasets for further processing.

From Table 3, it becomes apparent that no single query system is able to satisfy all the information requirements from literature. The most popular tool, the GBNV, only supports frequency plots and n-gram filtering, as do several others. Thus, they are unable to satisfy any frequency-related information requirements. For the systems that support frequency filtering, the actual capabilities vary: The *Ngram Search Engine* supports only frequency thresholds (Sekine and Dalwani 2010), while *Slash/A* (Todorova et al. 2014) can highlight time periods with significant fluctuations in frequency.⁹ CHQL supports different, more complex computations, for example nearest-neighbor search for frequency similarity and aggregation over arbitrary sets of n-gram frequencies (Willkomm et al. 2018). In addition, CHQL also supports information requirements from the NLP group, for example sentiment analysis. The only other tool that supports NLP information requirements is *NgramQuery*, by defining operators connected to *WordNet* (Fellbaum 1998; Miller 1995). The other systems differ mostly within individual groups of information requirements, for example with regards to their flexibility in wildcard usage or the exact frequency filtering predicates they allow.

Of the existing query systems, CHQL supports the most categories of information requirements, offering operators for every category except frequency plots. Another system notable for its flexible query language is *NgramQuery*: While it supports only three categories of information requirements, its NLP operators are more powerful than those of the other systems. However, neither of the two systems has a publicly available implementation (Richter et al. 2025), so researchers cannot actually use the systems for their studies or extend them for specific use cases. The other available systems, on the other hand, lack the required flexibility to express complex research questions from literature, as shown by the lack of support for different groups of information requirements.

5 Conclusions

5.1 Summary

Our work was motivated by three research questions, we summarize our answers as follows:

⁹<http://linguistics.chrisculy.net/lx/vistola/tools/slasha.html>

- **Q1:** What kind of (implicit or explicit) information requirements do researchers have when working with the GBNC?
 - The information requirements are diverse, but fall into six main groups: *frequency plots* (in 35 publications), *n-gram filtering* (52), *frequency filtering* (4), *external knowledge* (22), *NLP* (10), and *statistical computations* (30).
- **Q2:** Are existing query systems for the GBNC powerful enough to satisfy these information requirements?
 - While most groups of information requirements are supported by at least some of the systems, no system supports them all. The most powerful systems have no available implementations.
- **Q3:** If not, what is missing in these query systems?
 - Most existing tools offer n-gram filtering and frequency plots, but support for more complex analyses or the inclusion of external knowledge is very limited.

We answered **Q1** with an extensive literature review in Section 3. We identified four categories of research works, according to their usage of TNCs: *corpus as data*, *analysis*, *query systems*, and *corpus construction*. Across these paper categories, we extracted six main groups of information requirements, namely *frequency plots*, *n-gram filtering*, *frequency filtering*, *external knowledge*, *NLP*, and *statistical computations*. We then discussed these categories’ individual contributions. We have shown that information requirements are mostly domain-independent, so a TNC query system supporting these information requirements can be useful for researchers from many different fields.

To answer **Q2** and **Q3**, we researched additional existing query systems for TNCs and compared their capabilities to the information requirements we identified. Notably, most query systems are limited to plotting the frequencies of specific n-grams, with varying levels of wildcard support for n-gram filtering. None of them satisfy all of the information requirements. The two that come the closest, namely *NgramQuery* (Aleksandrov and Strapparava 2012) and *CHQL* (Willkomm et al. 2018; Schmidt-Petri et al. 2021), have no available implementations.

5.2 Future Work

From the answers we found to our research questions, one direction of future work is immediately apparent: A flexible, domain-independent TNC query system is theoretically possible, but does not exist yet. Publishing such a system could benefit researchers from many different disciplines.

Another direction of future work is the extension of the survey itself. As discussed in Section 3.1.1, we employed a relatively restrictive, title-based filter to publications. A broader survey may be possible in the future, e.g., by using *citation intent classification* (Cohan et al. 2019) instead of a pure keyword selection. Furthermore, the restriction to publications written in English potentially skews the findings towards Anglophone perspectives, making the inclusion of other languages a valuable extension of our work.

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